ECE 501 Project Week 8 Report

Project Title: Vehicle Orientation Estimation Using Conventional Image Processing

Group Number: 14

Project number: 8

Team Members:

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Github Repository Link: https://github.com/hetpatel5107/DIP_2026_Project8_Group14

- **Introduction**

In this week, we focused on comparing the performance of PCA and Radon for vehicle orientation estimation. We worked upon the results from 240 vehicle crops across five aerial images and saw how estimation errors vary with flight height, vehicle type, and ground-truth orientation. The aim was to understand where each method performs better and identify possible reasons behind the differences.

- **Flight Height Vs Error**

  Code:

```
import pandas as pd

import matplotlib.pyplot as plt



pca = pd.read_csv("Week7_PCA_Results.csv")

radon = pd.read_csv("week6_radon_results.csv")



heights = {

    "0Ago_SddH_jFbn": 66.25,

    "0av2_F0GS_ESe2": 63.11,

    "0B5X_l5jk_Fh2I": 92.91,

    "0bGO_UHlP_zpWU": 47.09,

    "0bL4_wk8I_7GPe": 65.88

}



pca["Image"] = pca["Source Image"].str.replace(".jpg", "", regex=False)

radon["Image"] = radon["Image"].str.replace(".jpg", "", regex=False)



results = pd.DataFrame({
```

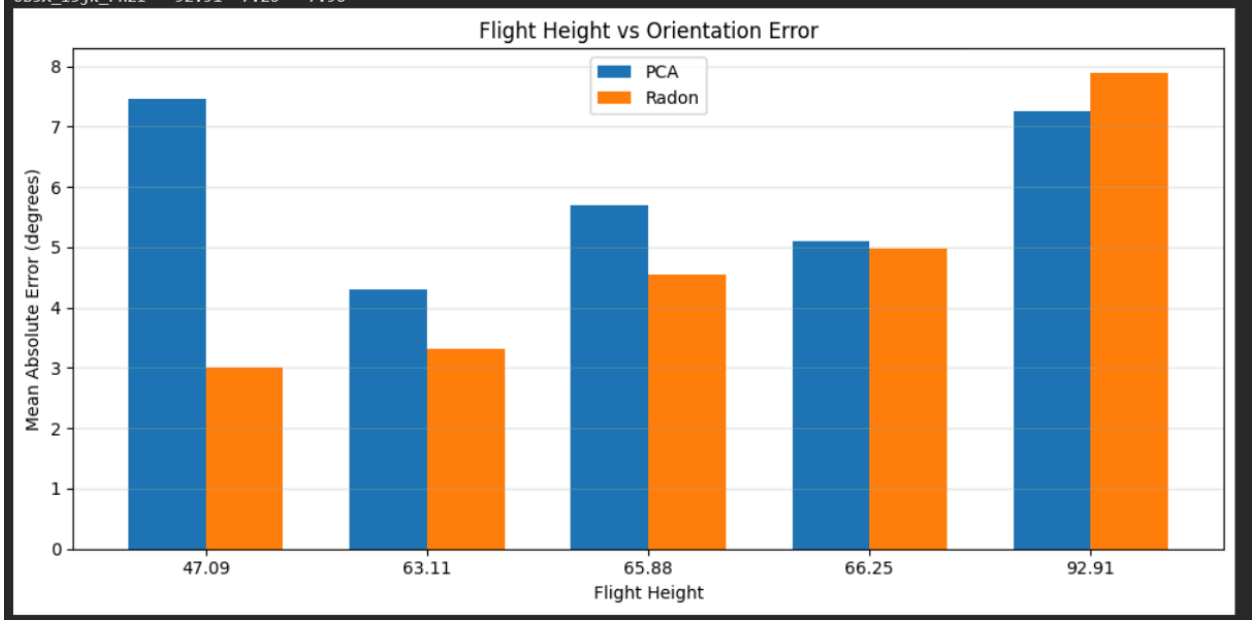
```
    "Height": pd.Series(heights),
    "PCA": pca.groupby("Image")["Error (degrees)"].mean(),
    "Radon": radon.groupby("Image")["Radon Error"].mean()
}).sort_values("Height")


print(results.round(2))


results[["PCA", "Radon"]].plot(kind="bar", figsize=(10, 5), width=0.7)
plt.xticks(range(5), [f"{h:.2f}" for h in results["Height"]], rotation=0)
plt.xlabel("Flight Height")
plt.ylabel("Mean Absolute Error (degrees)")
plt.title("Flight Height vs Orientation Error")
plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```

Output:

	Height	PCA	Radon
0bGO_UH1P_zpWU	47.09	7.47	3.00
0av2_F0GS_ESe2	63.11	4.30	3.31
0bL4_wk8I_7GPe	65.88	5.71	4.54
0Ago_SddH_jFbn	66.25	5.10	4.97
0B5X_15jk_Fh2I	92.91	7.26	7.90



Interpretation :

As shown in the graph above, the Radon error generally increases as the Flight Height increases, going from 3.00° at 47.09 to 7.90° at 92.91. Higher flight heights cause vehicles to cover less pixels resulting in less defined edges. Linear features are also less visible. The Radon method finds the dominant peaks within image projections to find orientation so the lack of edges will cause a less accurate estimation. Whereas, the PCA error varies. This is due to the PCA method using the general distribution of edge points to estimate orientation using covariance. Therefore the Radon and PCA methods are affected differently by the lack of details within images as Flight Height increases.

- **Vehicle Type Vs Error**

Code:-

```
class_errors = comparison_df.groupby("Type").agg(
    PCA=("PCA Error", "mean"),
    Radon=("Radon Error", "mean"),
    Count=("PCA Error", "count")
)

class_errors = class_errors.sort_values("PCA")

main_classes = class_errors[class_errors["Count"] >= 5]
rare_classes = class_errors[class_errors["Count"] < 5]

x = np.arange(len(main_classes))
width = 0.35

plt.figure(figsize=(12, 6))

plt.bar(
    x - width/2,
    main_classes["PCA"],
    width,
    label="PCA"
)

plt.bar(
    x + width/2,
    main_classes["Radon"],
    width,
    label="Radon"
)

plt.xlabel("Vehicle Type")
plt.ylabel("Mean Absolute Error (degrees)")
plt.title("PCA vs Radon Error by Vehicle Class")
```

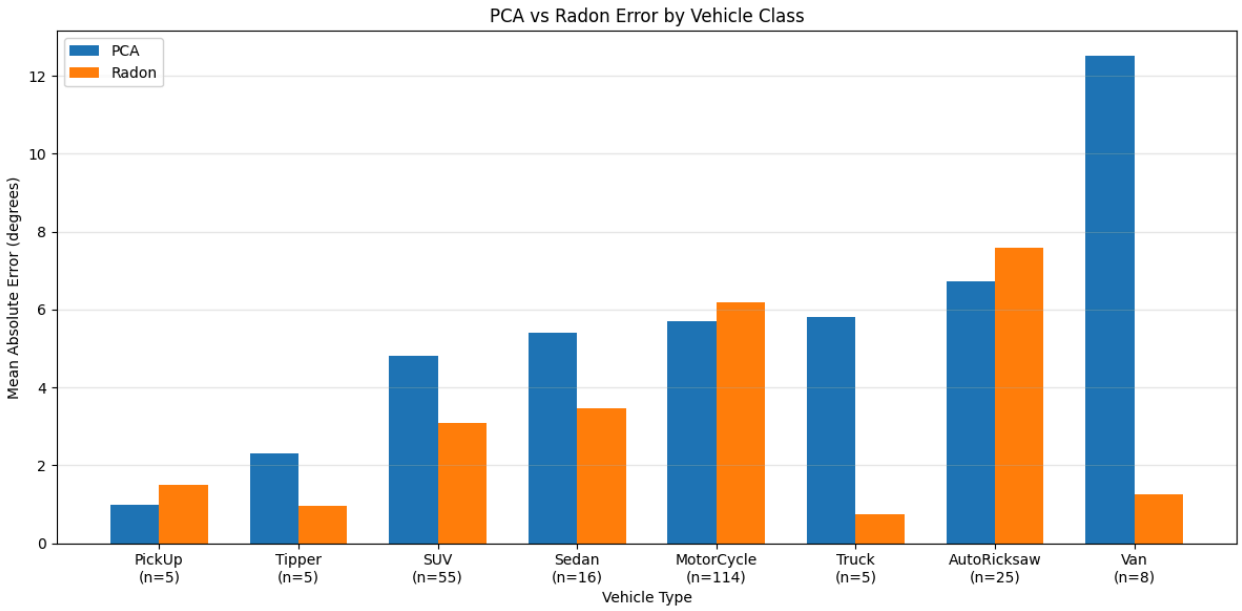
```
plt.xticks(
    x,
    [f"{name}\n(n={count})" for name, count in
     zip(main_classes.index, main_classes["Count"])]
)

plt.legend()
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

print("Vehicle Class Statistics:")
print(class_errors.round(2).to_string())

print("\nClasses with fewer than 5 samples:")
print(rare_classes.round(2).to_string())
```

Output:-



Vehicle Class Statistics:

Type	PCA	Radon	Count
PickUp	0.97	1.49	5

Tanker	1.94	0.36	2
Tipper	2.31	0.95	5
Auto3WCargo	3.94	1.60	1
Container	4.52	0.66	2
SUV	4.82	3.09	55
Bus	5.16	1.20	1
Sedan	5.40	3.47	16
MotorCycle	5.69	6.18	114
Truck	5.80	0.73	5
AutoRicksaw	6.73	7.58	25
Trailer	7.47	0.02	1
Van	12.52	1.26	8

Classes with fewer than 5 samples:

Type	PCA	Radon	Count
Tanker	1.94	0.36	2
Auto3WCargo	3.94	1.60	1
Container	4.52	0.66	2
Bus	5.16	1.20	1
Trailer	7.47	0.02	1

Interpretation:-

The plot of the orientation error for Vans reveals that there is a much bigger difference between the two methods: here, the Radon method is a great improvement over PCA (1.26 against 12.52). This difference is due to the fact that Vans have unique continuous rectangular boundaries. The Radon transform computes the sum of the intensities of these straight lines, and forms a large mathematical peak which is easily recognizable even in the presence of background noise to identify the true heading. In contrast, PCA chooses the principal axis according to the spatial variance of all the edge pixels in the entire image. Extraneous edge points mathematically will spread the PCA eigenvector away from the actual axis of the vehicle, if a Van's edge is affected by harsh shadows or background interference. This large performance margin needs to be

interpreted with caution, however, because the data set is small, comprising only 8 Van samples, with a few noisy outlier crops having a significant effect on the average class performance.

- **Ground truth Orientation Vs Error**

Code:

```
angle_bins = [0, 15, 30, 45, 60, 75, 90]

angle_labels = [

    "0-15°",

    "15-30°",

    "30-45°",

    "45-60°",

    "60-75°",

    "75-90°"

]



comparison_df["Angle Range"] = pd.cut(

    comparison_df["Ground Truth Axis (0-90)"],

    bins=angle_bins,

    labels=angle_labels,

    include_lowest=True

)



angle_errors = comparison_df.groupby(

    "Angle Range", observed=False
```

```
).agg(
    PCA=("PCA Error", "mean"),
    Radon=("Radon Error", "mean"),
    Count=("PCA Error", "count")
)

x = np.arange(len(angle_errors))
width = 0.35

plt.figure(figsize=(10, 6))

plt.bar(
    x - width/2,
    angle_errors["PCA"],
    width,
    label="PCA"
)

plt.bar(
    x + width/2,
    angle_errors["Radon"],
    width,
    label="Radon"
)
```

```
plt.xlabel("Ground Truth Vehicle Orientation")

plt.ylabel("Mean Absolute Error (degrees)")

plt.title("PCA vs Radon Error Across Orientation Ranges")



plt.xticks(

    x,

    [f"{name}\n(n={count})" for name, count in

     zip(angle_errors.index, angle_errors["Count"])]

)



plt.legend()

plt.grid(axis="y", alpha=0.3)

plt.tight_layout()

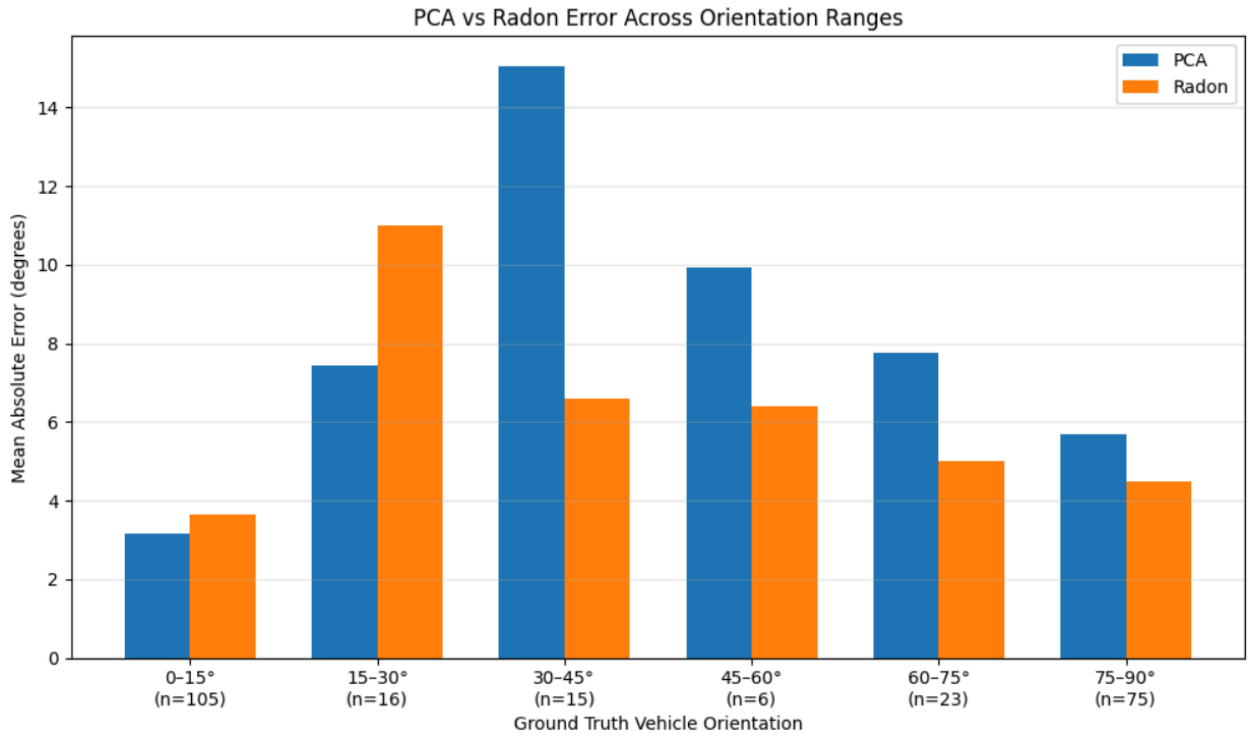
plt.show()



print("Orientation-wise Error Statistics:")

print(angle_errors.round(2).to_string())
```

Output:



Orientation-wise Error Statistics:

PCA Radon Count

Angle Range

0–15° 3.17 3.65 105

15–30° 7.43 11.00 16

30–45° 15.07 6.61 15

45–60° 9.92 6.39 6

60–75°	7.77	5.02	23
75–90°	5.69	4.50	75

Interpretation:

Figure 2 plots the average orientation error values for PCA and Radon algorithms over six ground truth angle intervals. Both algorithms exhibit low errors within the 0°-15° interval – 3.17° and 3.65°, respectively. The maximum PCA error of 15.07° is observed within the 30°-45° interval while the Radon algorithm shows a lower value of 6.61°. This discrepancy can be explained by the uneven visibility of vehicle edges that influences PCA's ability to determine the dominant orientation of the vehicle. PCA determines the orientation of the vehicle based on the overall distribution of the image pixels, and Radon determines it based on the cumulative projection lines. Only 15 samples are present within the 30°-45° interval in contrast to 105 and 75 within 0°-15° and 75°-90°, respectively. Therefore, the average value of this interval is less reliable. In general, Radon demonstrates lower mean values of errors than PCA.

- **Conclusion**

The comparison helped us understand that PCA and Radon perform differently depending on the image conditions and vehicle characteristics. Neither method consistently performs better in every situation. Radon's reliance on prominent linear features and PCA's dependence on the distribution of edge points provide possible explanations for these differences. Overall, this analysis gave us a better understanding of the strengths and limitations of both methods.

1. Work done this week

- Combined PCA and Radon results for all 240 vehicle crops.
- Analysed the relationship between flight height and orientation estimation error.
- Compared errors across different vehicle classes.
- Studied the effect of ground-truth orientation on estimation accuracy.
- Created plots to visualise the comparison.

2. Work to be done next week

- Review and finalise the results from all experiments.
- Complete the final project report.
- Prepare the final project presentation.

3. Challenges faced

- Understanding the differences in PCA and Radon performance across flight heights and vehicle classes.
- Explaining why some orientation ranges produced higher errors.
- Finding theoretical reasons to support the observations rather than relying only on numerical comparisons.